Technical Supplement: Bilingual Skill Routing at 80K Scale

Anonymous submission

Public Benchmark Construction

The SkillRouter Eval Core release (Zheng et al. 2026) contains 87 tasks and two candidate pools. We use its 79,141-record hard tier and omit two source-excluded and 12 `generic_only` tasks, leaving 73 tasks and 182 distinct core gold skill identifiers. Every gold identifier is present in the candidate pool.

The released multilingual file stores the source fields and three query columns: exact English, Simplified Chinese, and the same Chinese task with one fixed bilingual routing instruction. The first translation pass was generated by one isolated language-model run. A second isolated run compared every translated row with the exact English source and corrected all 73 rows. The audit focused on omitted steps, negations, numeric constraints, paths, code symbols, fenced code blocks, and output schemas. A deterministic validator then checks:

exact English preservation, complete task coverage without duplicates, Chinese-script presence in translated queries, both Chinese and Latin scripts in instruction-prefixed queries, exact prefix equality, and the absence of email-shaped personal identifiers.

The data card and main paper disclose that these are not professional human translations.

Candidate Representation

For all methods, the candidate text contains only the skill name and description. The public records also provide a body, but using it would change the research question from pre-activation metadata routing to full-text routing. Word and character methods join fields with a period. Generic multilingual-E5 receives `passage: name. description`. The SkillRouter-domain encoder receives `name | description`, matching the field delimiter in its public model card while omitting the body.

Hyperparameters

Word TF–IDF uses 1–2 grams, sublinear term frequency, and a 300K feature cap. BM25 uses unigrams, $k_1 = 1.2$, $b = 0.75$, and a 300K cap. Character TF–IDF uses 2–5 grams, minimum document frequency 2, and a 300K cap. mE5-small uses 384 dimensions, length 512, and batch 128; SR-Emb uses 1024 dimensions, length 512, and batch 8. RRF combines the top 20 SR and character results with constant 60. Bootstrap confidence intervals use 10,000 samples and seed 20270827. No task-specific model training is performed. Both Hugging Face models are pinned to exact 40-character revisions in `config/model_revisions.json`; cache metadata additionally binds the model revision, ordered skill identifiers, formatted document text, matrix shape, and output checksum. Exactly one value was tried for each public-benchmark setting; values were fixed before evaluation from standard formulations, model cards, and the source benchmark protocol rather than selected on test performance.

The public dataset and model caches are downloaded or generated once. Query latency includes query vectorization or model encoding plus exact search over the cached candidate matrix. Sparse index fitting and dense cache load-and-validation time are recorded as `setup_seconds`. Offline candidate embedding generation is a separate one-time step and is not timed in the reported result. Equal scores use ascending skill identifier as a stable secondary key. The public SkillRouter dataset is pinned to an immutable Hugging Face commit in `config/dataset_revisions.json`, and every downloaded file is checked against a committed SHA-256 digest.

Reproduction

The release includes:

- SHA-256 checksums and a downloader for the public SkillRouter hard tier;
- an audited multilingual JSONL file and deterministic validator;
- exact dependency locks;
- scripts for both embedding caches and all retrieval methods;
- result JSON, rankings, figures, and task-level error analysis; and
- a Makefile that compiles the paper, checklist, supplement, and archives.

The full workflow is:

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```
make setup
make data validate-data
make cache-e5 cache-skillrouter
make experiments test
make submission
```

The reported run used a 14-core Apple M4 Pro with 64 GB memory, Apple MPS, macOS 26.6.1, Python 3.12.14, PyTorch 2.13.0, scikit-learn 1.9.0, and sentence-transformers 5.7.0. Exact versions are stored in `requirements-aaai-lock.txt`.

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The source task text is derived from SkillsBench (Li et al. 2026), whose repository uses Apache License 2.0, and is redistributed by SkillRouter Eval Core. The multilingual variants are marked as modifications and retain upstream attribution. The large candidate pool and model caches are not committed; the release downloads them from their official locations. Code written for this study uses the MIT License. Third-party skill descriptions retain their original terms.

References

Li, X.; Liu, Y.; Chen, W.; et al. 2026. SkillsBench: Benchmarking How Well Agent Skills Work Across Diverse Tasks. arXiv:2602.12670.

Zheng, Y.; Zhang, Z.; Ma, C.; Yu, Y.; Zhu, J.; Wu, Y.; Xu, T.; Dong, B.; Zhu, H.; Huang, R.; and Yu, G. 2026. SkillRouter: Skill Routing for LLM Agents at Scale. arXiv:2603.22455.